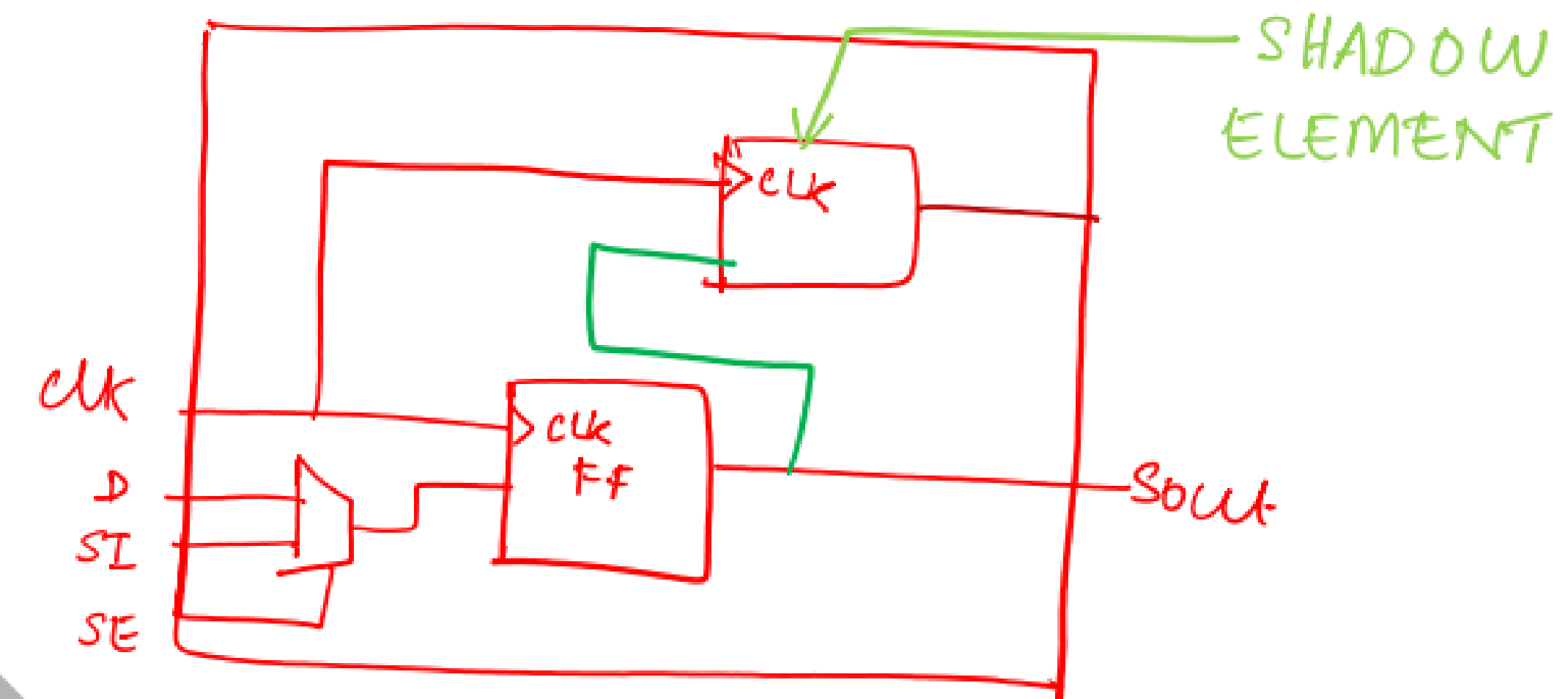
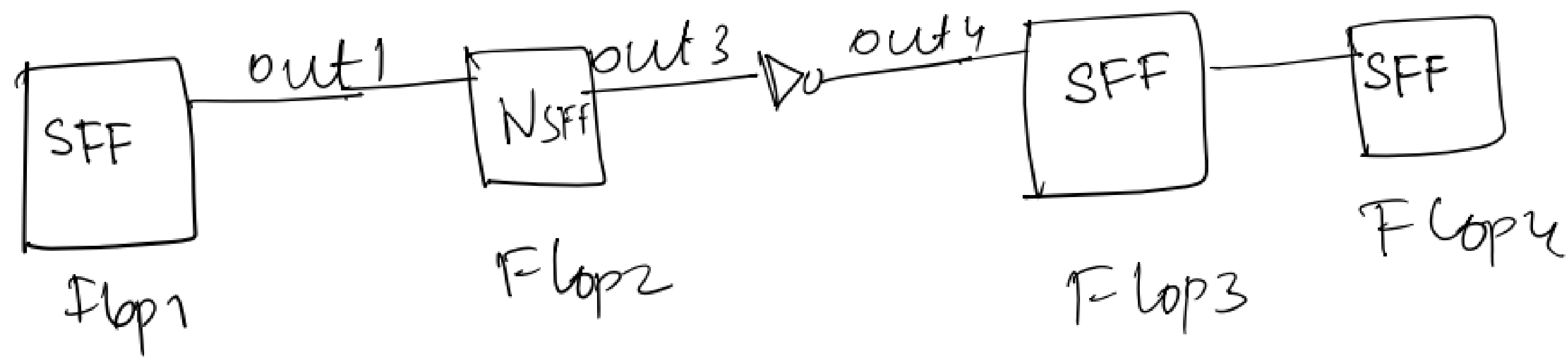




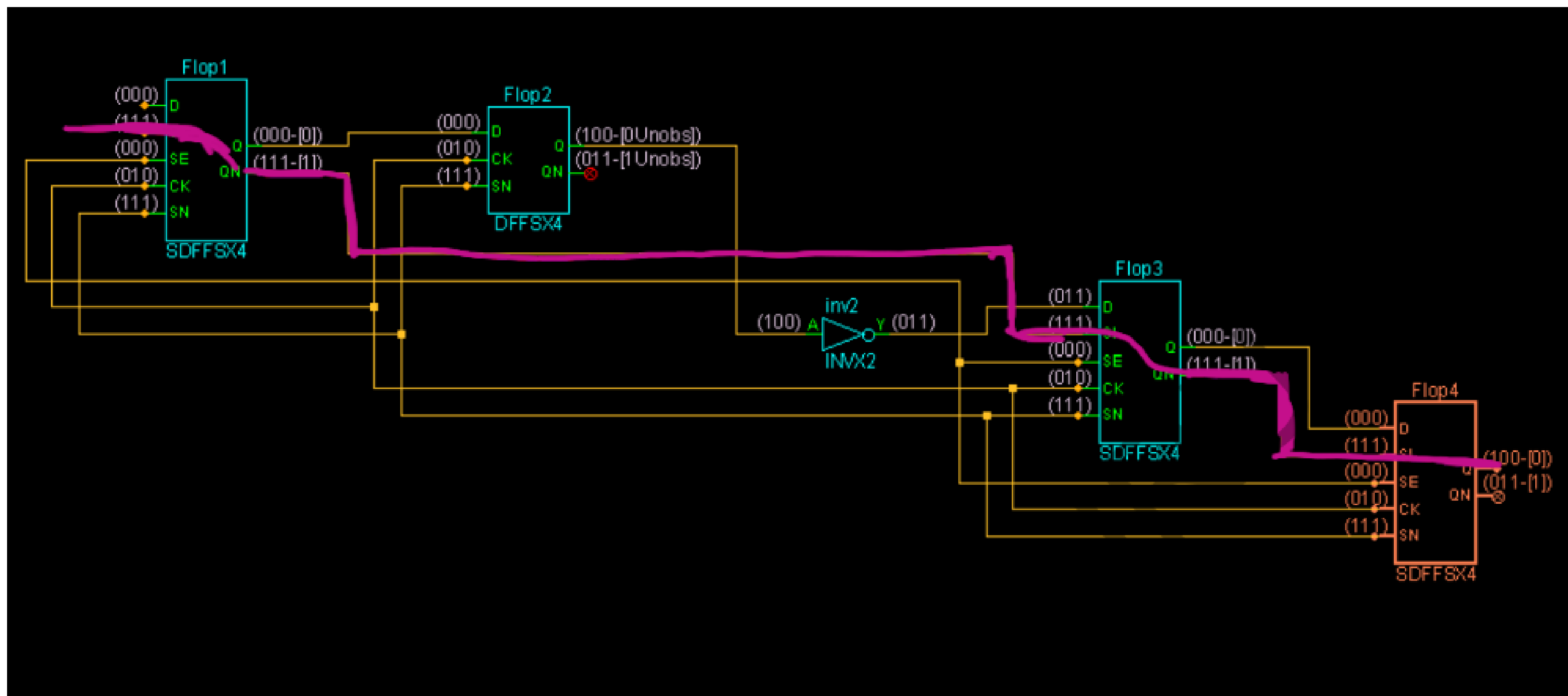
```
module try (out, FastClk, Reset, d);
input FastClk, Reset, d;
output out;
wire out_1, out_3, out_4, out_5;
```

```
DFFSX4 Flop1 ( .D(d), .CK(FastClk), .SN(Reset), .Q(out_1) );
DFFSX4 Flop2 ( .D(out_1), .CK(FastClk), .SN(Reset), .Q(out_3) );
INVX2 inv2 ( .A(out_3), .Y(out_4) );
DFFSX4 Flop3 ( .D(out_4), .CK(FastClk), .SN(Reset), .Q(out_5) );
DFFSX4 Flop4 ( .D(out_5), .CK(FastClk), .SN(Reset), .Q(out) );
```

```
endmodule
```

```
set_context dft -scan
read_verilog try.v
read_cell_library /hdd2/home/vidishar/feb23/library/tsmc13.mdt
set_current_design try
add_clocks 0 FastClk
add_clocks 1 Reset
add_nonscan_instance Flop2
set_system_mode analysis
report_drc_rules
insert_test_logic -number 1
write_netlist outputs/try_scan.v -replace
write_atpg_setup outputs/atpg -replace
```

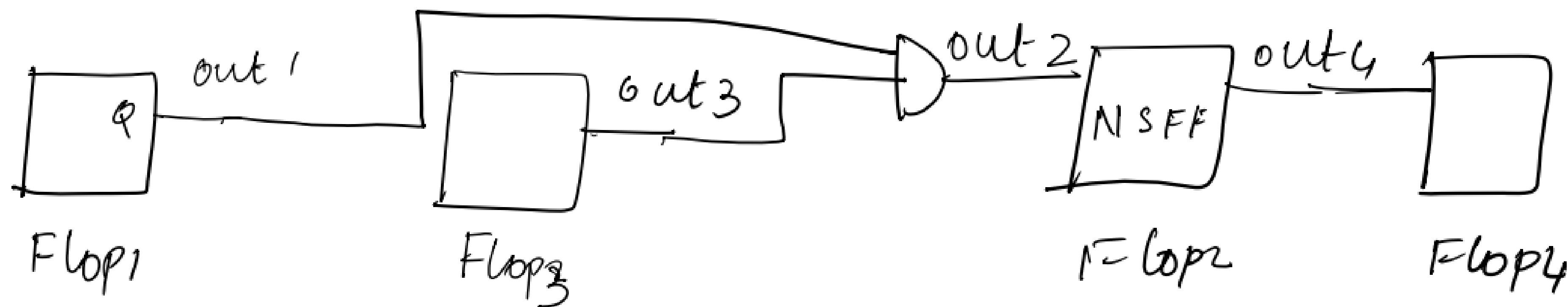
If a scan flop's o/p is directly connected to the i/p of non scan ff $\Rightarrow$ shadow element



NOTE : Shadow element is NOT a part of scan chain. The tool just mentions it in scan cell report. The scan chain is highlighted in the fig. As seen, flop2 is not a part of scan chain.

cell#	chain	memory_type	inv	gate#	shift_clock	inv	cell_name	instance_name	(ext. pin names)
0	chain1	MASTER	(FF-LE)	FFFF	26	/FastClk	F	SDFFSX4	/Flop4 I2 (SI,Q)
1	chain1	MASTER	(FF-LE)	TFFT	25	/FastClk	F	SDFFSX4	/Flop3 I2 (SI,QN)
		SHADOW	(FF-LE)	TFTF	24	/FastClk	F	DFFSX4	/Flop2 I1 (-,-)
2	chain1	MASTER	(FF-LE)	FTFT	23	/FastClk	F	SDFFSX4	/Flop1 I2 (SI,QN)

If it is a shadow element, then the tool can predict the value of the non-scan ff during shift phase because the O/p of scan ff is directly connected to the i/p of non scan ff.



```

module try (out, FastClk, Reset, d1, d2);
input FastClk, Reset, d1, d2;
output out;
wire out_1, out_2, out_3, out_4;

```

```

DFFSX4 Flop1 ( .D(d1), .CK(FastClk), .SN(Reset), .Q(out_1) );
DFFSXL Flop3 ( .D(d2), .CK(FastClk), .SN(Reset), .Q(out_3) );
AND2XL and1 (.Y(out_2), .A(out_1), .B(out_3));
DFFSX4 Flop2 ( .D(out_2), .CK(FastClk), .SN(Reset), .Q(out_4) );
DFFSX4 Flop4 ( .D(out_4), .CK(FastClk), .SN(Reset), .Q(out) );

```

```

endmodule
~
~

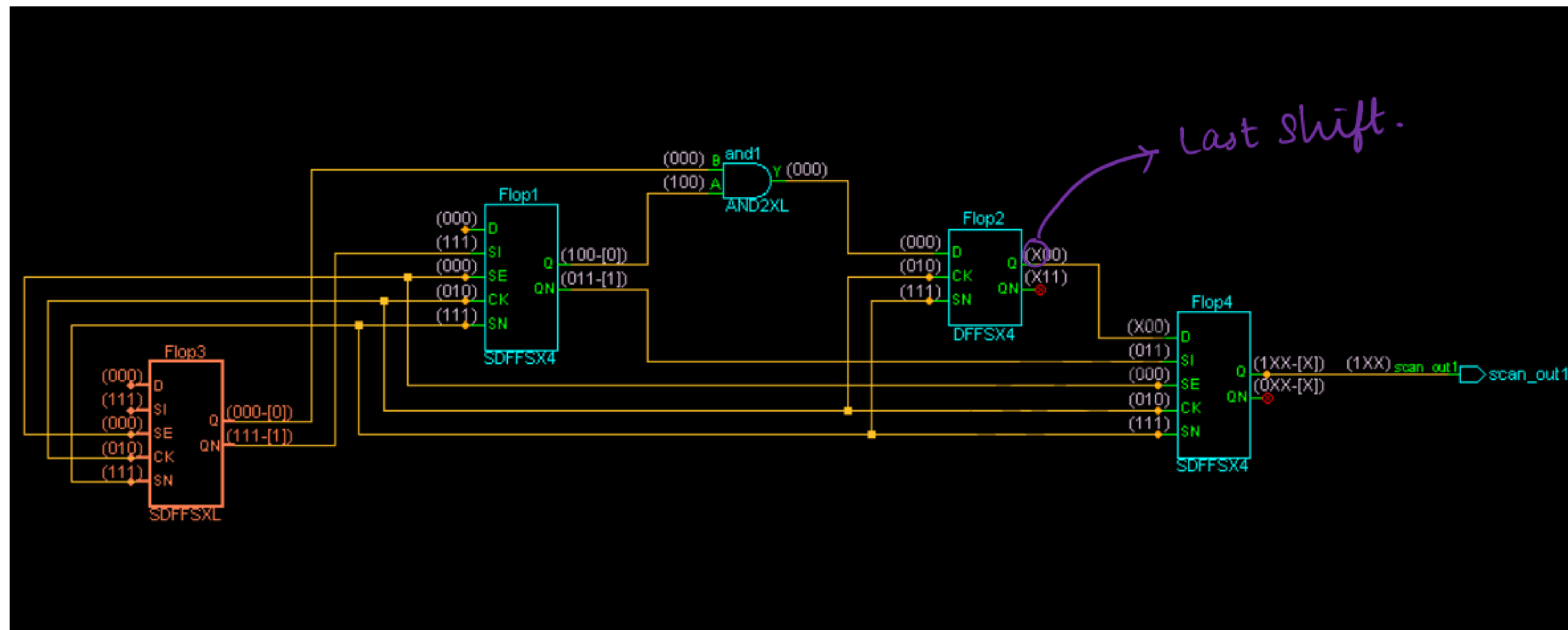
```

```

set_context dft -scan
read_verilog try.v
read_cell_library /hdd2/home/vidishar/feb23/library/tsmc13.mdt
set_current_design try
add_clocks 0 FastClk
add_clocks 1 Reset
add_nonscan_instance Flop2
set_system_mode analysis
report_drc_rules
insert_test_logic -number 1
write_netlist outputs/try_scan.v -replace
write_atpg_setup outputs/atpg -replace

```

If a scan ff's o/p through a combo, if it gets connected to the i/p of non scan ff
 ⇒ NOT A shadow cell.



cell#	chain	memory_type	inv	gate#	shift_clock	inv	cell_name	instance_name	(ext. pin names)
0	chain1	MASTER	(FF-LE)	FFFF	29	/FastClk	F	SDFFSX4	/Flop4 I2 (SI,Q)
1	chain1	MASTER	(FF-LE)	TFFT	26	/FastClk	F	SDFFSX4	/Flop1 I2 (SI,QN)
2	chain1	MASTER	(FF-LE)	FTFT	27	/FastClk	F	SDFFSXL	/Flop3 I2 (SI,QN)

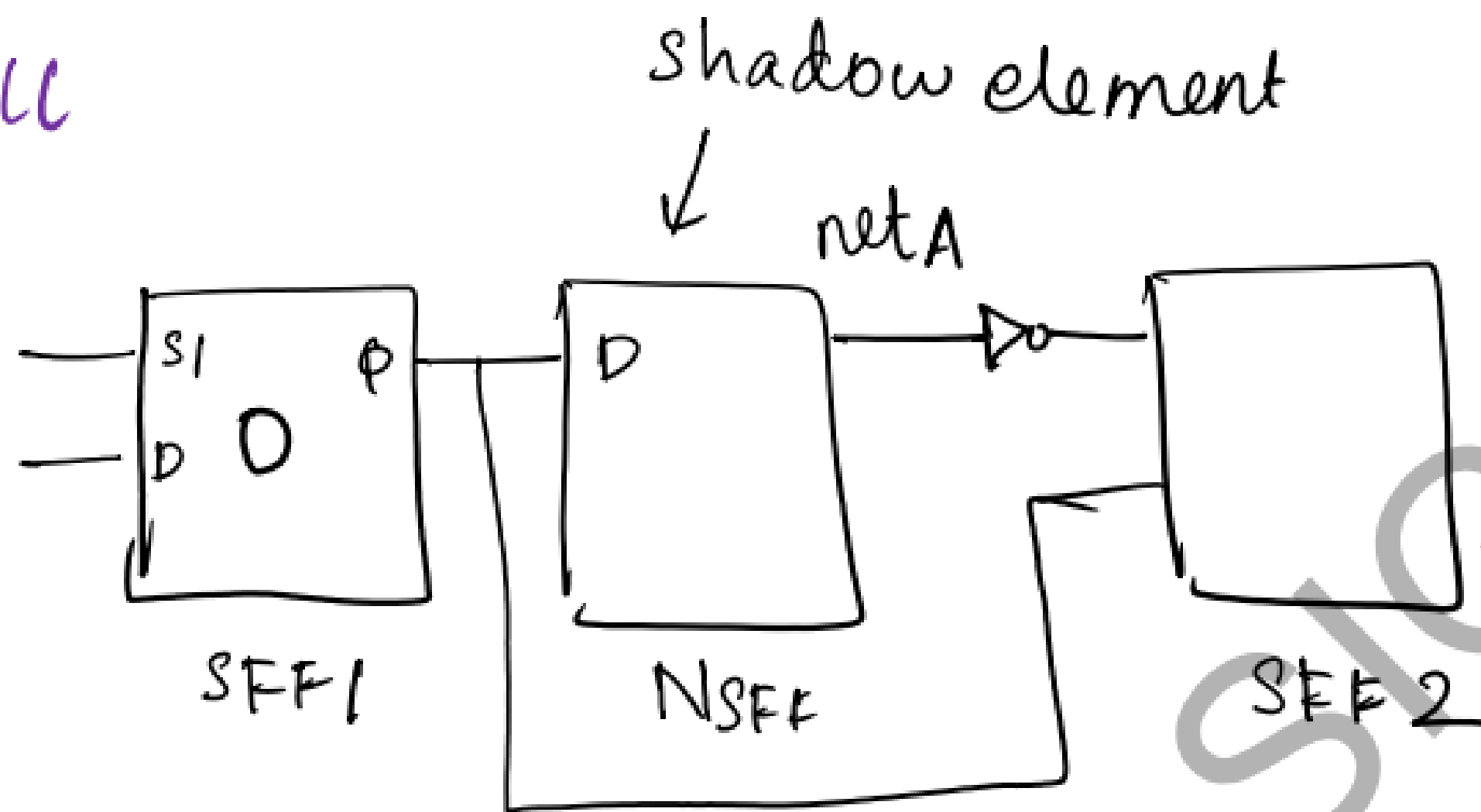
Here during shift phase, the tool can't predict the value of nonscan ff because the scanff's o/p through combo is connected to the i/p of non scan ff. Tool will not analyze combo during shift (tool's algo is like that) $\therefore$ It's o/p is x during shift

Parallel fail serial pass scenario.

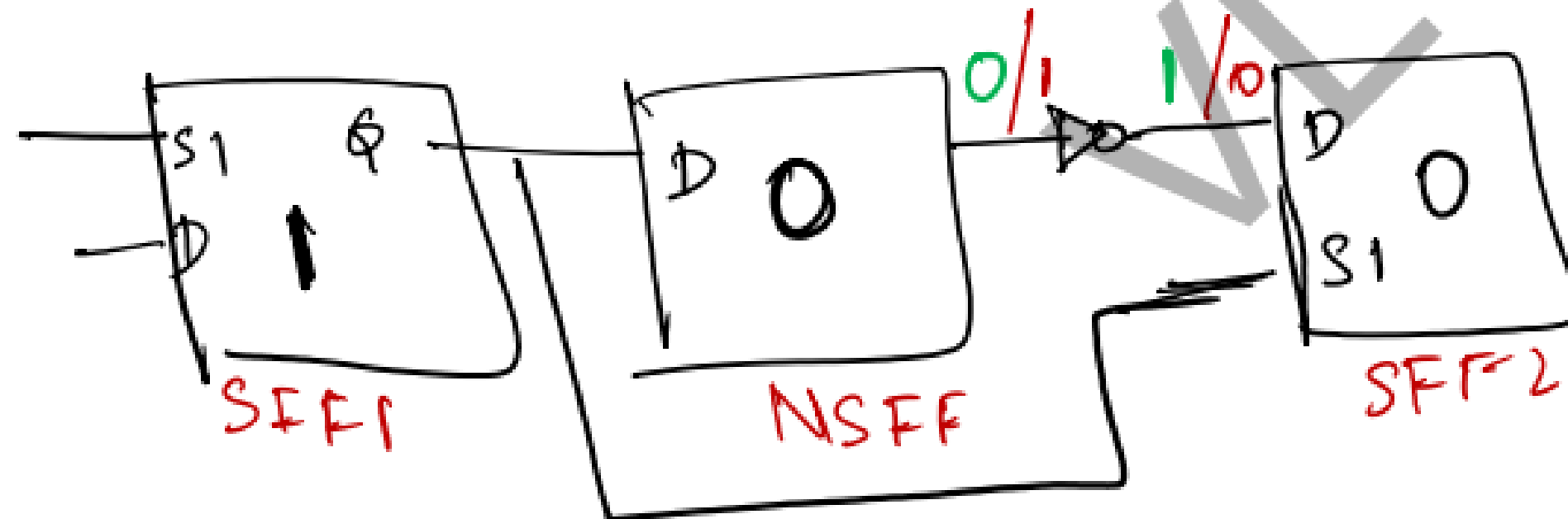
Why does shadow cell results in parallel simulation mismatch?

ATPG tool while writing pattern assumes that the shadow element will get the previous value of scan ff. And that value will be applied to combo.

ATPG tool will think like this when shadow element is there.



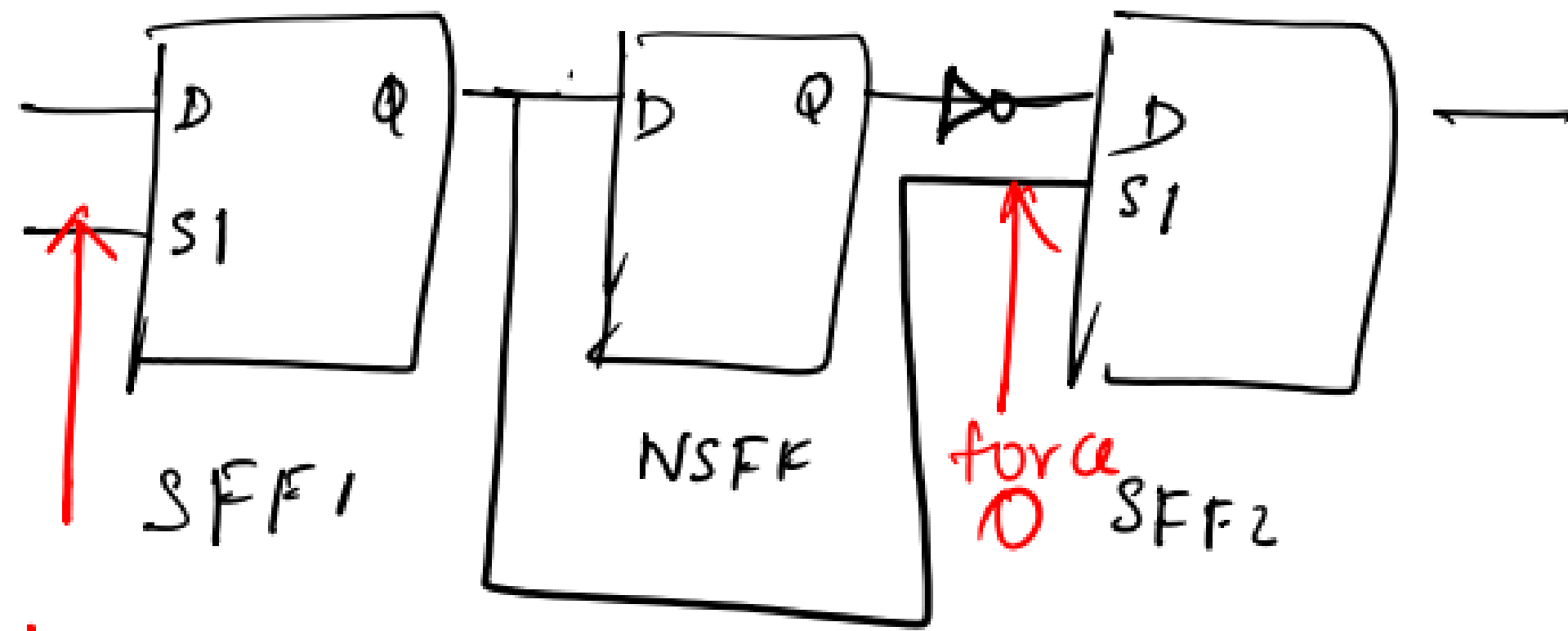
Eg:- To detect SAI on net A.
Pattern will be shifted:- 10
(as in this eg there are 2 ffs in scan chain $\Rightarrow$ SFF1 & SFF2)



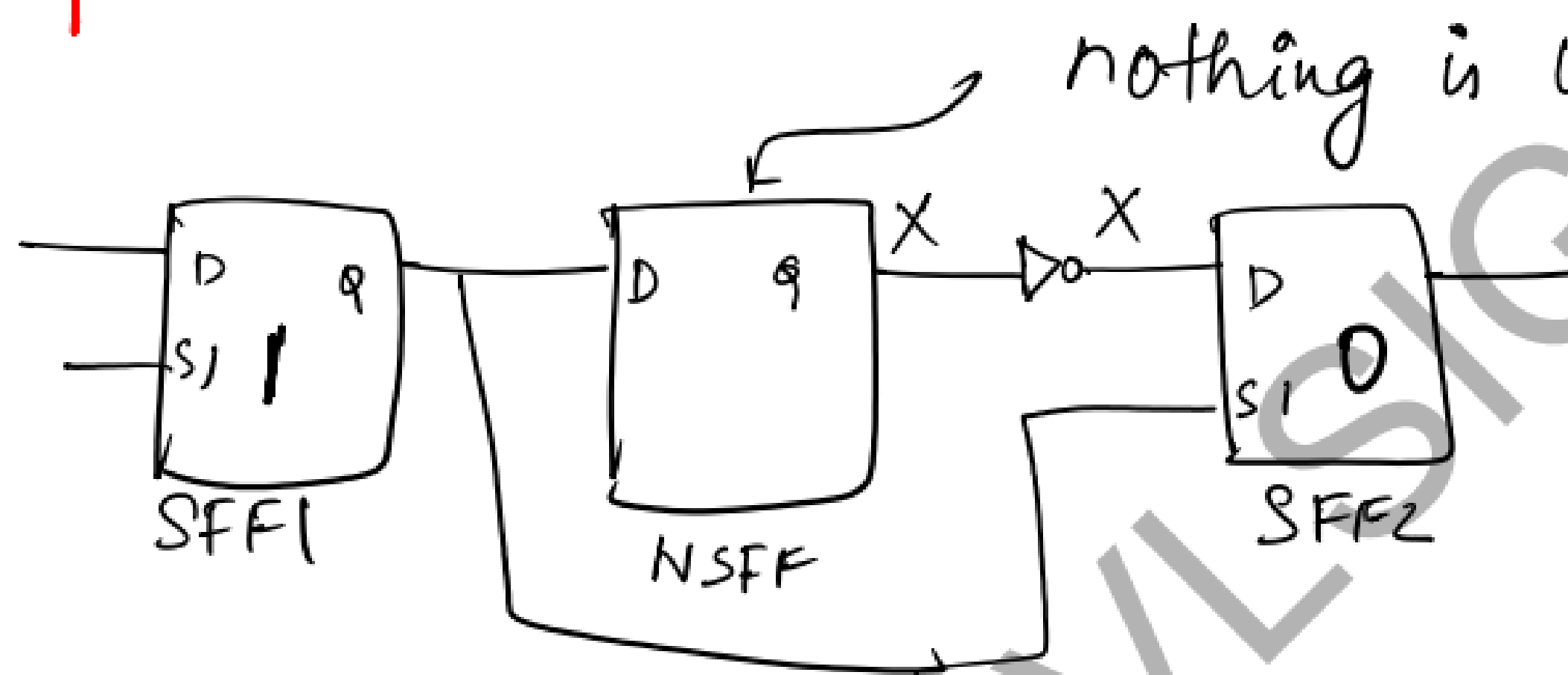
Capture :- Good m/c $\Rightarrow$ SFF2 should capture 1
faulty m/c $\Rightarrow$ SFF2 should capture 0

} $\Rightarrow$ Expected value :-
SFF2 should capture 1

But during 11th simulation, we load the data 11th into the scan bff
 pattern :- 10



force 1



nothing is loaded into shadow element
 (Mentor Graphics)

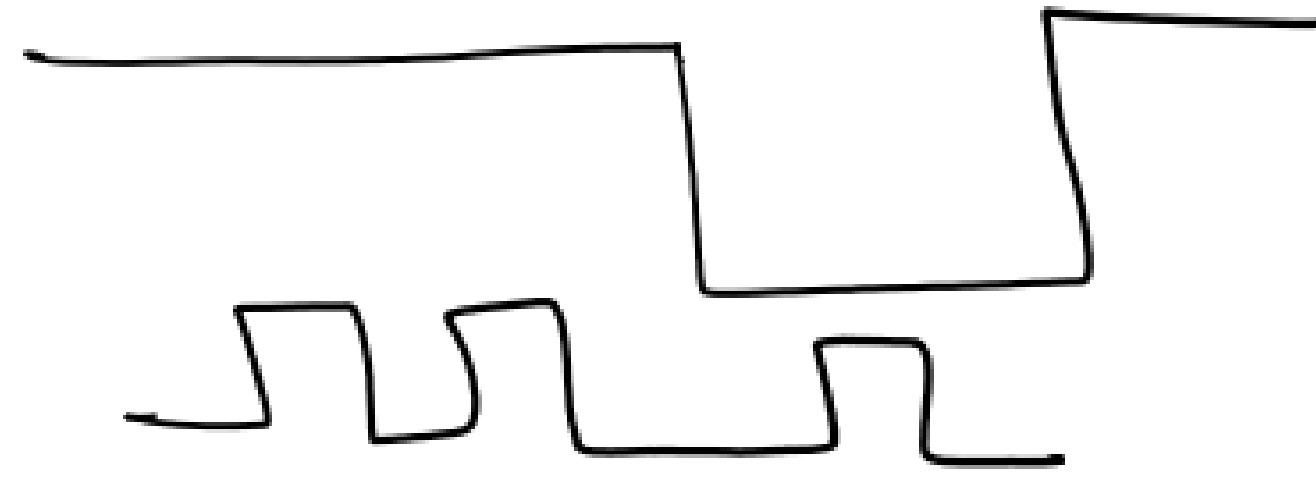
Given in Tenent Official website:

Correct loading of shadow cells depends on shift activity. Because parallel patterns lack the requisite shift activity to load shadow cells correctly, you may get simulation mismatches with parallel patterns that disappear when you use serial patterns.

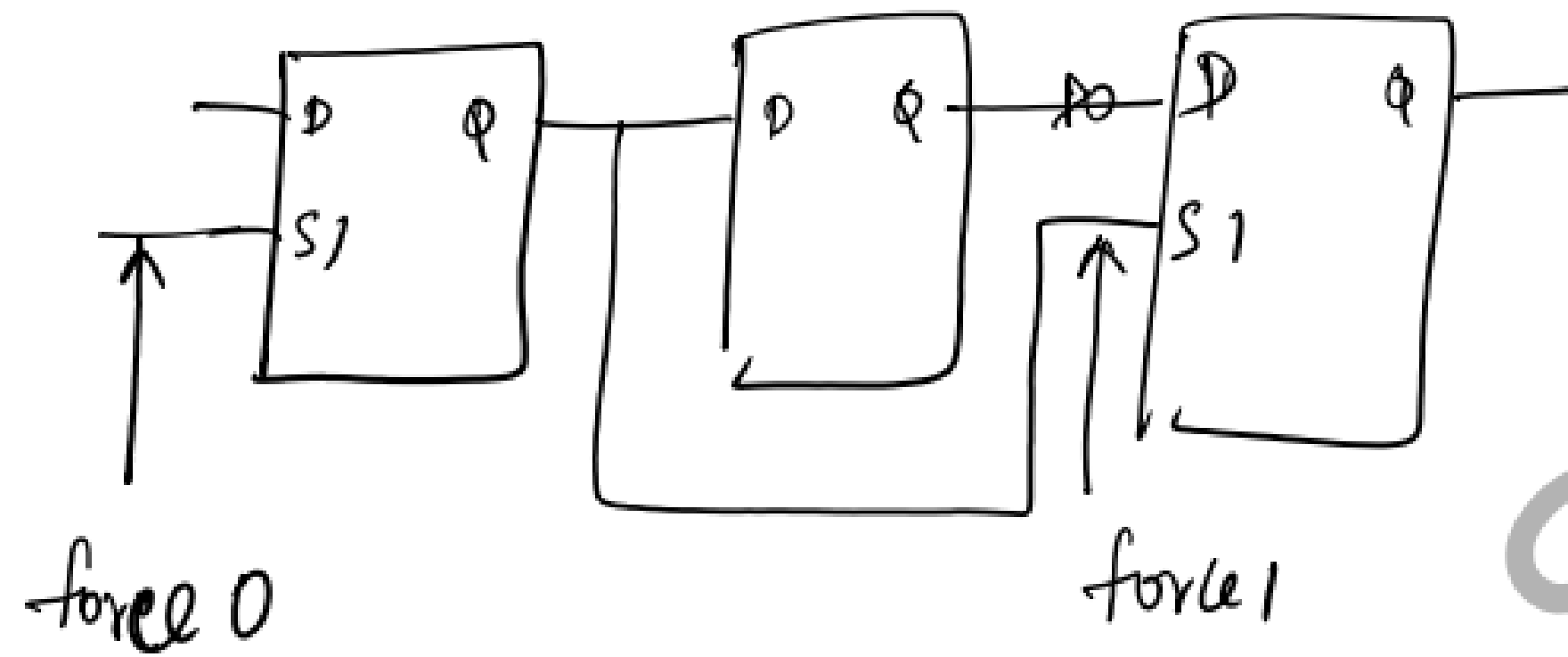
capture :- $\therefore$ SFF2 will capture X.

patterns that disappear when you use serial patterns.

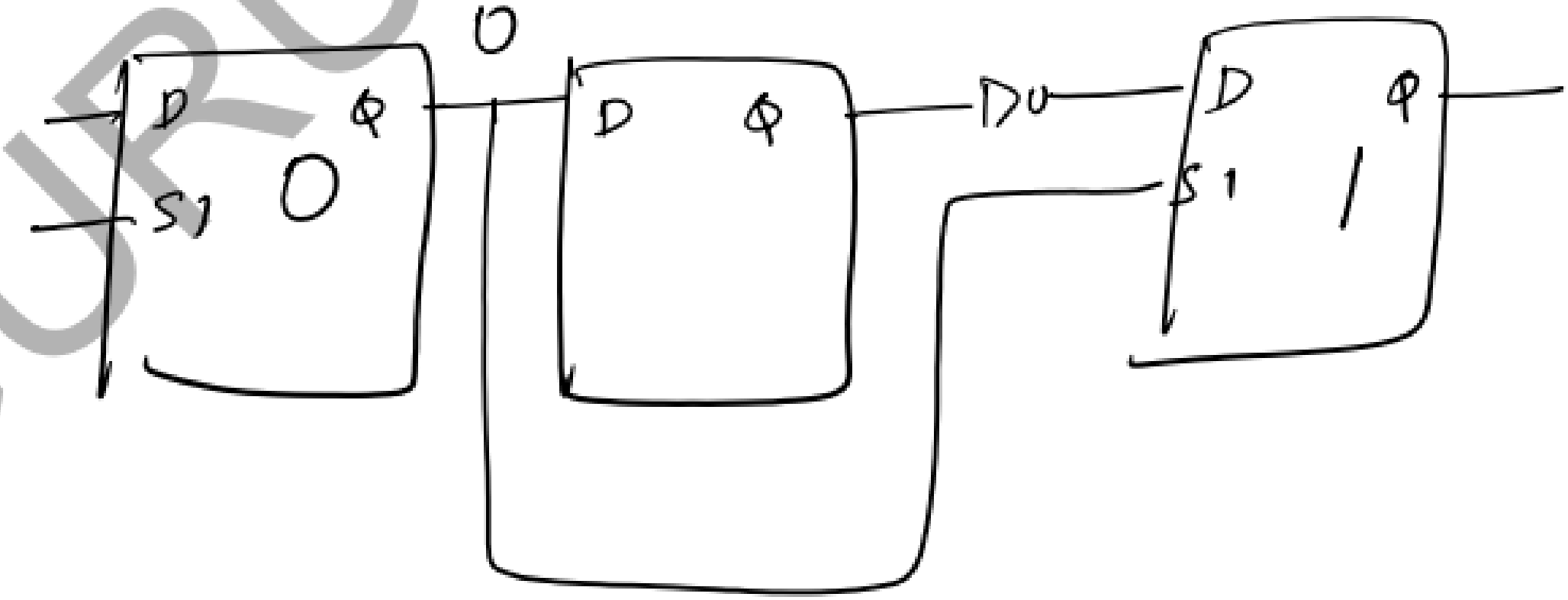
Tessent tool solves this problem by giving 2 pulses when $SEN=1$ when there is shadow element
 (default:- in 112 sim, only 1 pulse will be there when $SEN=1$)



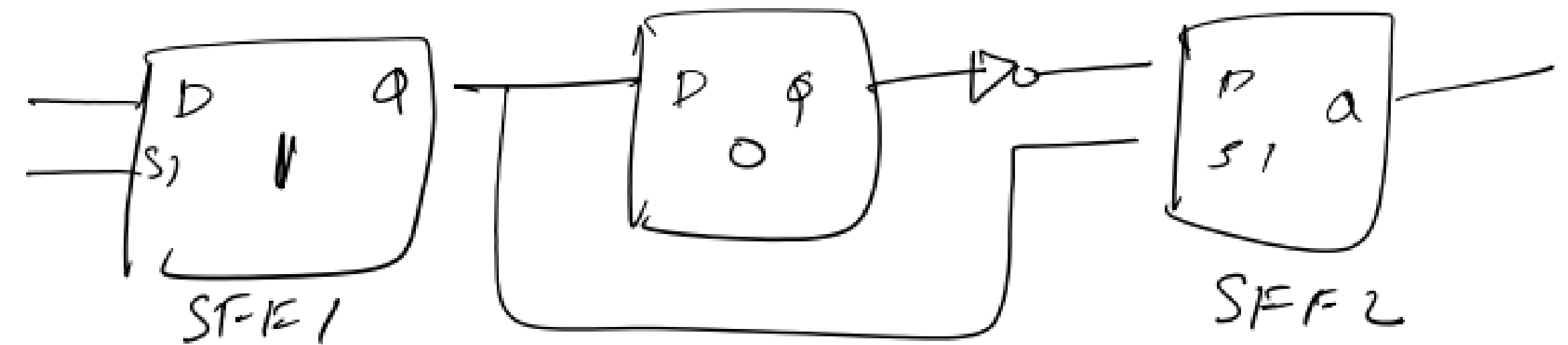
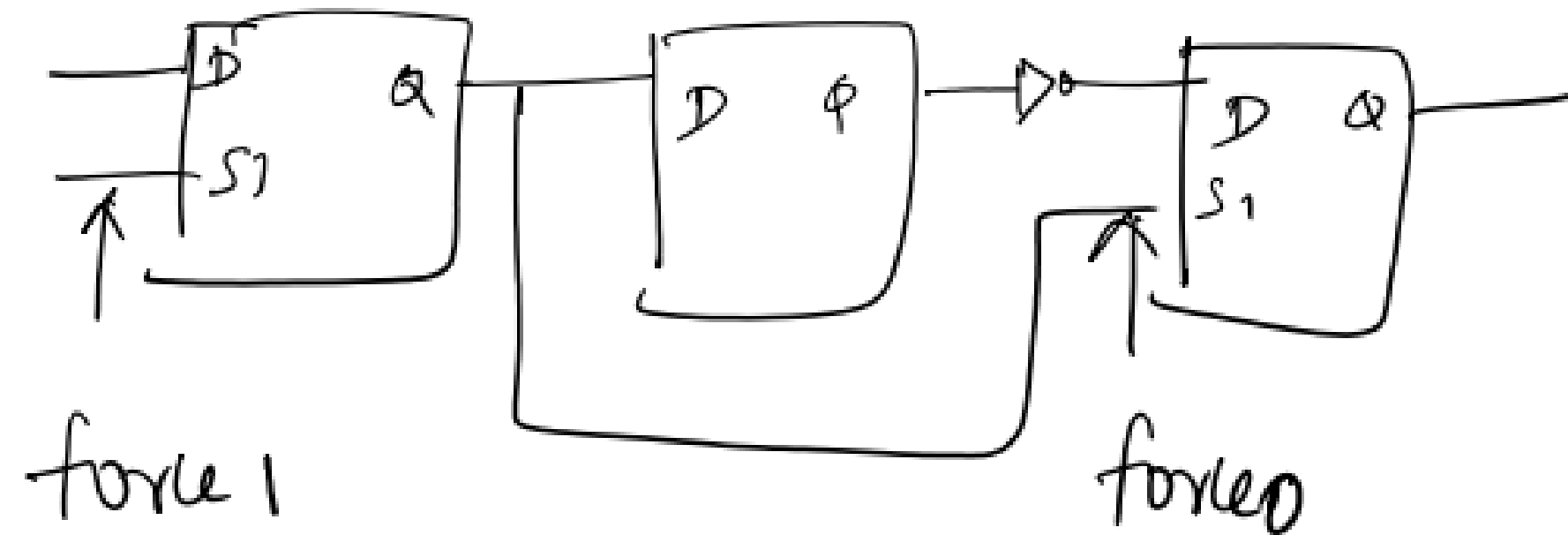
eg:-



pulse clk



pulse clk

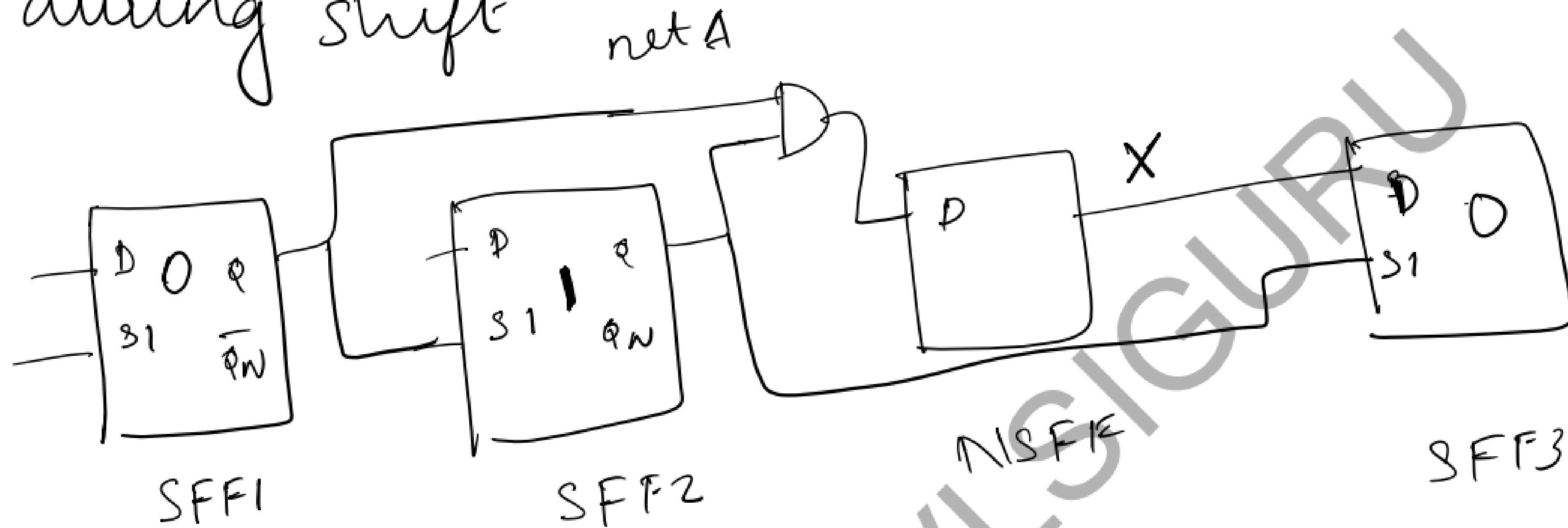


capture:- SFF2 $\Rightarrow$ 1

Why non scan is not causing mismatches?

Because for non-scan, ATPG will assume its value as X during shift

To detect SA1 on net A.



In this case, we need 2 pulses in capture (Seq depth concept)

capture :- 1st pulse SFF3 $\Rightarrow$ X

2nd pulse SFF3 $\Rightarrow$

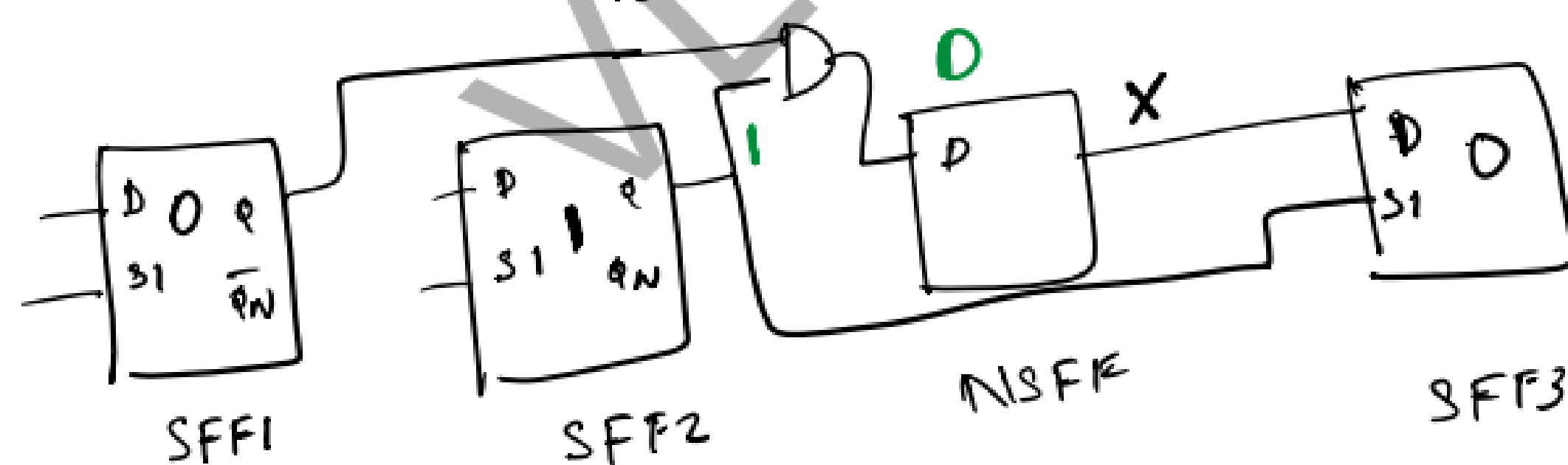
g	t
0	1

 Expected value:-
SFF3 $\rightarrow$ after 2nd capture pulse = 0

SE

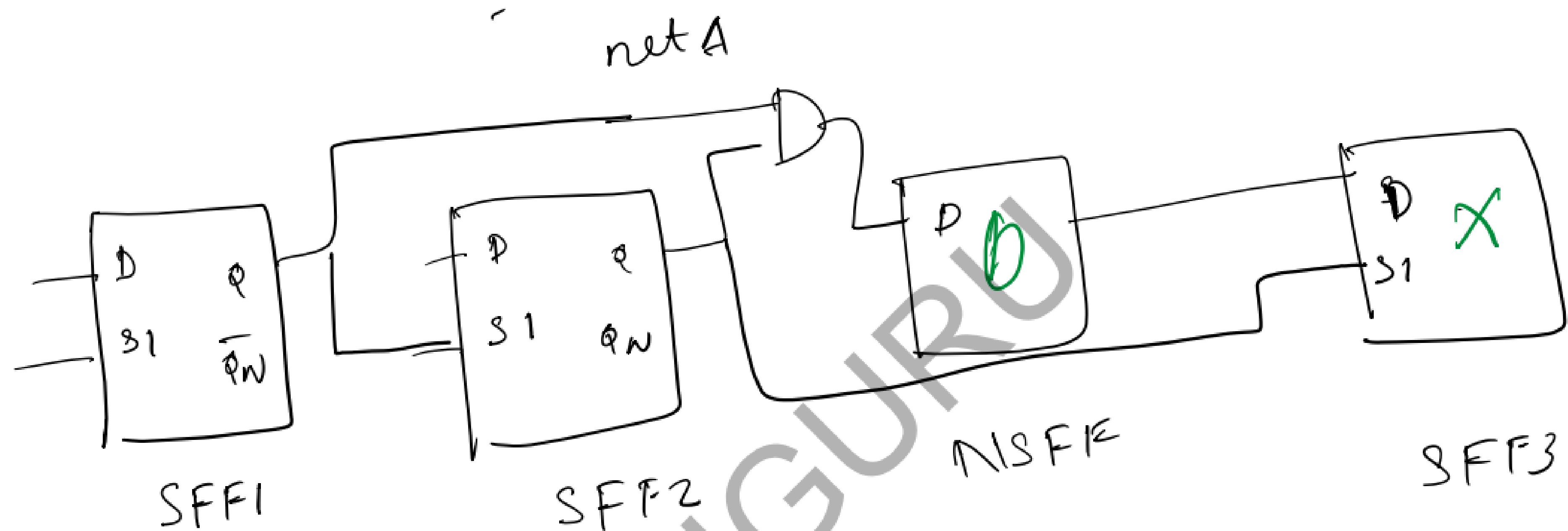


$\delta_{net A}$

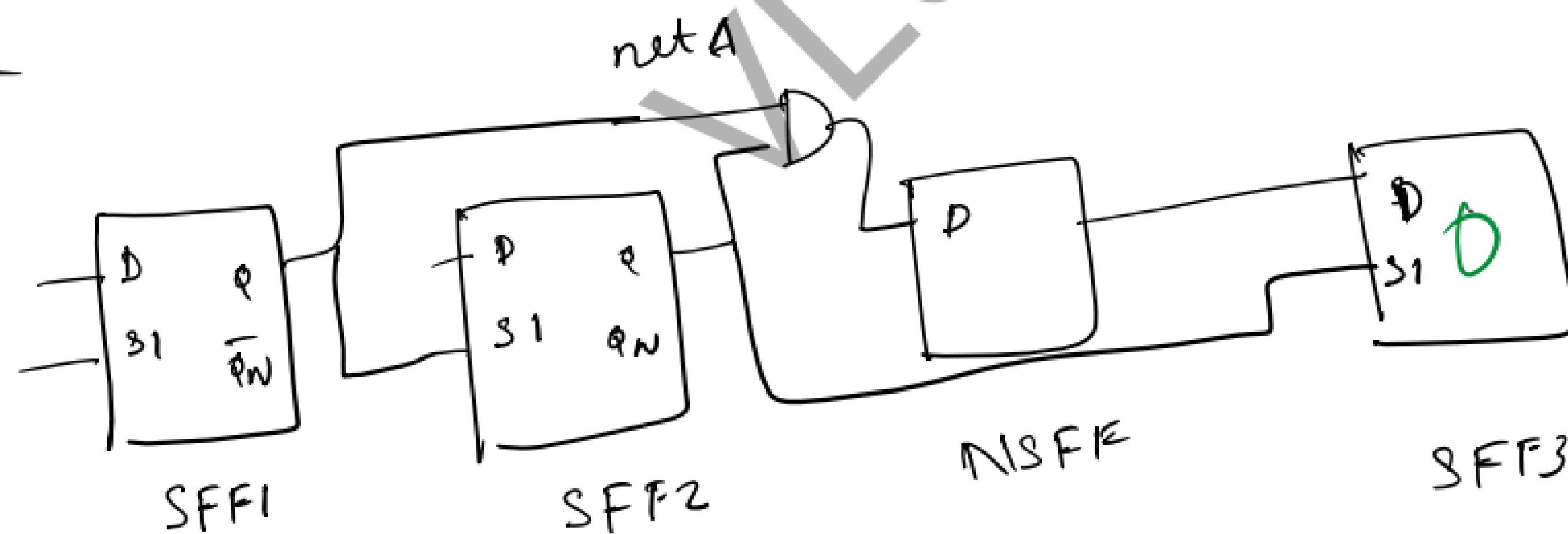


Capture

1st pulse



2nd pulse



Problem in Shadow element :- ATPG tool predicts that shadow element will take up ~~the~~ previous value of scan cell.

But in 11th sim, by default it is not possible as only 1 pulse is there in $SEN=1$

But essent tool overcomes this problem by giving 2 pulses when $SEN=1$.

Non-scan cell :- no problem as ATPG predicts in the same way as sim env.

Scenario where parallel passes but serial fails:

Parallel patterns simulate relatively quickly, but do not detect problems that occur when data is shifted through the scan chains. One such problem, for example, is data shifting through two cells on one clock cycle due to clock skew. Serial patterns can detect such problems.

VLSIGURU